

Pointers

Lab questions, set 4

31. Write a C program to demonstrate pointer increment.

Assume an integer array:

```
int a[5] = {10, 20, 30, 40, 50};
```

Use a pointer to traverse the array using p++ and print both the value and the address of each element.

32. Write a C program to traverse the following array in reverse order using pointer decrement:

```
int a[5] = {5, 15, 25, 35, 45};
```

33. Write a C program to access array elements using pointer arithmetic.

Assume:

```
int a[6] = {2, 4, 6, 8, 10, 12};
```

Access elements using expressions like *(p + i).

34. Write a C program where two pointers point to elements of the same array:

```
int a[8] = {11, 22, 33, 44, 55, 66, 77, 88};
```

Find the number of elements between the two pointers using pointer subtraction.

35. Write a C program to compare two pointers pointing to the same array:

```
int a[5] = {3, 6, 9, 12, 15};
```

Demonstrate the use of relational operators (<, >, ==) on pointers.

36. Write a C program to show that:

array name = base address of array

Assume:

```
int a[4] = {7, 14, 21, 28};
```

Print addresses using:

- i. a
- ii. &a[0]
- iii. pointer variable

37. Write a C program and comment on why the following operations are illegal:

- i. Adding two pointers
 - ii. Multiplying a pointer by a constant
 - iii. Assigning a value to an address
- Use suitable code snippets for explanation.

38. Write a C program to traverse and print a 2D array using a pointer to array.

Assume:

```
int a[3][4] = {  
    {15, 27, 11, 35},  
    {22, 19, 31, 17},  
    {31, 23, 14, 36}  
};
```

39. Write a C program to calculate the sum of all elements of the array using pointer arithmetic.

Assume:

```
int a[5] = {4, 8, 12, 16, 20};
```

40. Write a C program to find the largest element in an array using pointers only.

Assume:

```
int a[6] = {45, 12, 78, 34, 89, 23};
```